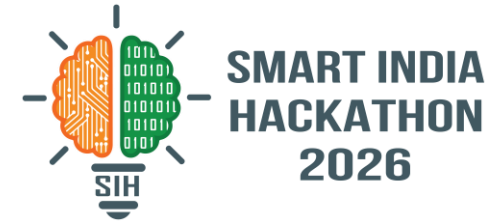


SMART INDIA HACKATHON 2026

IDEA PRESENTATION | SOFTWARE



BAGHEWALA / WELL-TO-SURFACE DIGITAL TWIN

INTEGRATED CSS + SRP DECISION SUPPORT

A running prototype that connects reservoir, wellbore and pump-system models for explainable monitoring and what-if analysis.

PROBLEM STATEMENT ID	[ADD EXACT ID FROM SIH PORTAL]
PROBLEM STATEMENT TITLE	Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field
THEME / CATEGORY	Smart Automation / Software
TEAM ID / REGISTERED TEAM NAME	[ADD TEAM ID] / [ADD REGISTERED TEAM NAME]

[LIVE DEMO | SIH-120-FINAL.VERCEL.APP | SIMULATED DATA](#)





[TEAM NAME]

IDEA TITLE



BAGHEWALA DIGITAL TWIN

Integrated CSS + SRP decision support for heavy-oil wells

THE PROBLEM | A COUPLED FAILURE CHAIN



PROPOSED SOLUTION

One shared well model

Connect reservoir temperature to estimated viscosity, wellbore conditions, rod/pump loading and production estimates.
Compare CSS and SRP scenarios in one operator workflow.

HOW IT ADDRESSES THE PROBLEM

From reactive to informed

Surface how reservoir cooling can affect viscous flow and pumping risk.
Show model alerts and compare options before a simulated change.

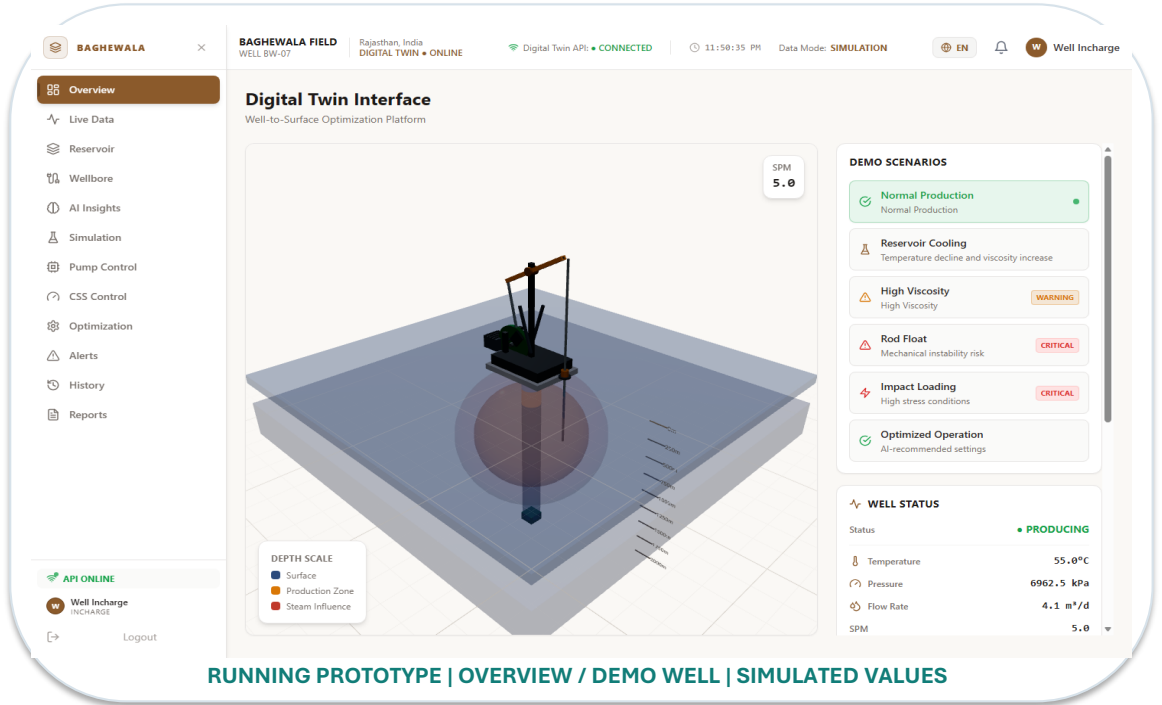
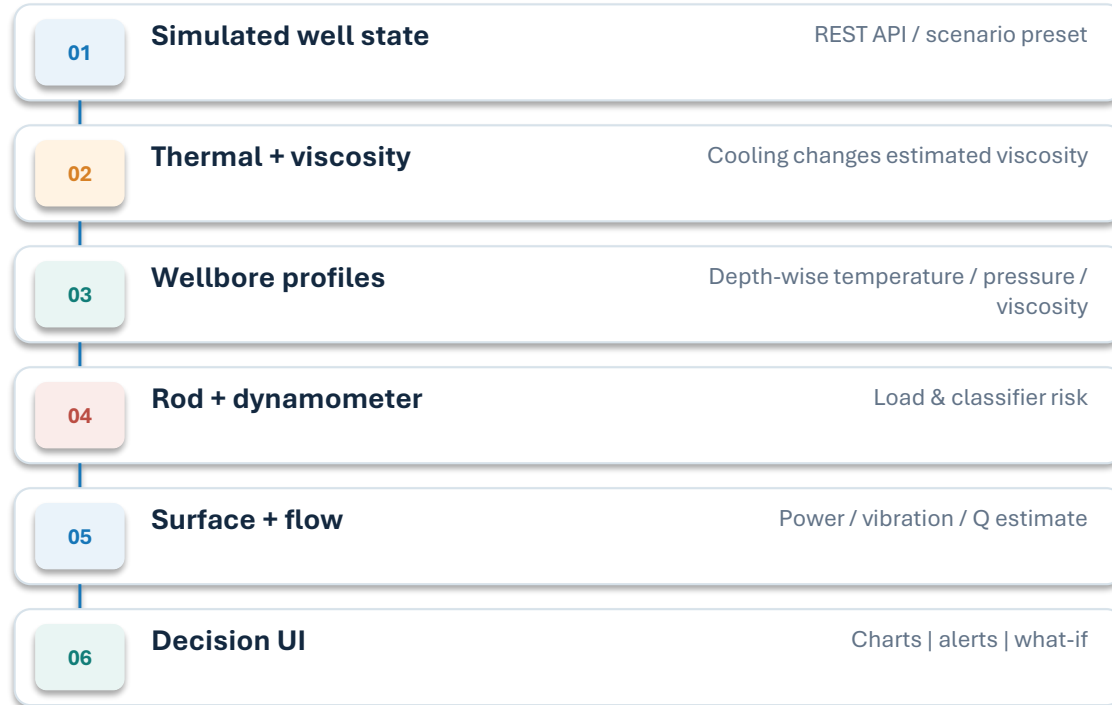
INNOVATION & UNIQUENESS

Integration + explainability

Cause-to-effect view across reservoir, wellbore and SRP instead of isolated dashboards.
What-if is separated from state-changing demo controls; engineer stays in control.

PROTOTYPE TODAY: simulated telemetry + simplified model + rule-based insights | FIELD AI / AUTO-CONTROL: future, only after data validation and safety approval

METHODOLOGY | THE RUNNING MODEL CHAIN



TECHNOLOGIES USED IN THE WORKING PROTOTYPE

React + TypeScript

Three.js

FastAPI + Python

NumPy

REST API

Zustand + Recharts

NEXT VALIDATION GATE

Field historian, calibrated models, shadow-mode advisory AI

Production deployment would require durable storage / streaming; Vercel demo state is not a field historian or control plane.

ANALYSIS OF FEASIBILITY

WORKING NOW

Deployable software demo

React + FastAPI prototype is live; model, dashboards and what-if endpoints run with simulated state.

DATA PATH

Field inputs are identified

PS lists production, CSS, steam, VFD/SRP, failure, completion, reservoir and fluid records.

VIABILITY GATE

Advisory before automation

Start offline, then shadow mode, then operator-approved advisory; no writes until formal safety approval.

POTENTIAL CHALLENGES & RISKS

Sparse / inconsistent records

Heavy-oil model mismatch

False alerts / operator overload

Cyber / operational safety

Demo state is not persistent

STRATEGIES TO OVERCOME THEM

Audit units, timestamps, well IDs; quantify missingness.

Calibrate by well / cycle; validate on held-out periods.

Tune thresholds; track precision, recall and false alarms.

Read-only shadow mode; access controls, interlocks, sign-off.

Add durable database + event stream before field deployment.

VIABILITY PATH | REPLAY HISTORICAL DATA -> CALIBRATE / VALIDATE -> SHADOW MODE -> APPROVED ADVISORY PILOT



[TEAM NAME]

IMPACT AND BENEFITS



POTENTIAL IMPACT ON TARGET AUDIENCE

PETROLEUM / RESERVOIR Understand thermal-fluid context	ARTIFICIAL LIFT / PRODUCTION Compare CSS + SRP decisions	FIELD OPERATORS See explainable, reviewable alerts	MAINTENANCE / ASSET Track reliability indicators
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BENEFITS OF THE SOLUTION | EXPECTED, NOT YET MEASURED

ECONOMIC Target: lower energy / barrel, fewer unplanned interventions, better value from steam.	OPERATIONAL Earlier visibility of changing thermal / pump conditions; more consistent decisions.
RELIABILITY Flag simulated rod-float / impact risk; future validation could support maintenance planning.	ENVIRONMENTAL If efficiency improves: potential reduction in steam and energy intensity per unit oil.

HOW A FIELD PILOT PROVES VALUE				
Oil rate / recovery	SOR	kWh per unit oil	Rod failures / unsets	Downtime
Compare to a verified baseline; publish only measured or validated results. No quantified production, SOR, energy or failure reduction is claimed.				



RESEARCH AND REFERENCES



RESEARCH BASIS | PROBLEM CONTEXT + MODEL FOUNDATIONS

PRIMARY PROBLEM SOURCE	Oil India Limited / SIH 2026 problem statement supplied by the team: heavy-oil Baghewala; CSS + SRP co-optimization; available production, cycle, steam, VFD/SRP, failure and reservoir/fluid data.	
CSS THERMAL DESIGN	Boberg, T.C. & Lantz, R.B. (1966). Calculation of the Production Rate of a Thermally Stimulated Well. JPT 18(12), 1613-1623.	Conceptual CSS reference; current prototype uses a simplified thermal model.
SRP DYNAMICS	Gibbs, S.G. (1963). Predicting the Behavior of Sucker-Rod Pumping Systems. JPT 15(7), 769-778.	Reference for rod-system modeling; current model is a simplified demonstration.
PHYSICS-INFORMED ML	Raissi, M., Perdikaris, P. & Karniadakis, G.E. (2019). Physics-informed neural networks. J. Comput. Phys. 378, 686-707. DOI: 10.1016/j.jcp.2018.10.045	Future research direction only; no trained PINN is in the running prototype.
IMPLEMENTATION DOCS	React: react.dev FastAPI: fastapi.tiangolo.com Vercel: vercel.com/docs	Frontend / API stack in the deployed demonstration.

[REPOSITORY / LIVE DEMO](#) | github.com/qwertypoiuy9/sih-120-final | sih-120-final.vercel.app | [Verify SIH portal PS ID and registered team details before submission.](#)